

Exercise J01

Q1. The chance that a 13 card combination from a pack of 52 playing cards is dealt to a player in a game of bridge, in which 9 cards are of the same suit, is

- (A) $\frac{4 \cdot {}^{13}C_9 \cdot {}^{39}C_4}{{}^{52}C_{13}}$ (B) $\frac{4! \cdot {}^{13}C_9 \cdot {}^{39}C_4}{{}^{52}C_{13}}$ (C) $\frac{{}^{13}C_9 \cdot {}^{39}C_4}{{}^{52}C_{13}}$ (D) None of these

Ans. A

Sol. -

Q2. One Indian and four American men and their wives are to be seated randomly around a circular table. Then the conditional probability that the Indian man is seated adjacent to his wife given that each American man is seated adjacent to his wife is

- (A) $1/2$ (B) $1/3$ (C) $2/5$ (D) $1/5$

Ans. C

Sol. -

Q3. If two different numbers are taken from the set $\{0, 1, 2, 3, \dots, 10\}$; then the probability that their sum as well as absolute difference are both multiple of 4, is

- (A) $\frac{14}{45}$ (B) $\frac{7}{55}$ (C) $\frac{6}{55}$ (D) $\frac{12}{55}$

Ans. C

Sol. $n(S) = {}^{11}C_2 = 55$
Both will be either from $\{2, 6, 10\}$ or $\{0, 4, 8\}$
Prob. = $\frac{{}^3C_2 + {}^3C_2}{55} = \frac{6}{55}$

Q4. Two dice are rolled simultaneously. The probability that the sum of the two numbers on the top faces will be atleast 10 is:

- (A) $1/6$ (B) $1/12$ (C) $1/18$ (D) None

Ans. A

Sol. -

Q5. A die is thrown. Let A be the event 'an odd number turns up' and B be the event 'a number divisible by 3 turns up'. Write which of the following are true.

- (i) $B \subset A$ (ii) $B - A \subset A$ (iii) $B - A \subset A^C$ (iv) $A - B = A \cap B^C$

- (A) (i) and (ii) (B) (i), (iii) (C) (ii) and (iii) (D) (iii) and (iv)

Ans. D

Sol. -

- Q6. Event A, B, C are pairwise mutually exclusive events such that $P(A) = \frac{3x+1}{3}$, $P(B) = \frac{1-x}{4}$ and $P(C) = \frac{1-2x}{2}$. The set of possible values of x are in the interval.
- (A) $[0, 1]$ (B) $\left[\frac{1}{3}, \frac{1}{2}\right]$ (C) $\left[\frac{1}{3}, \frac{2}{3}\right]$ (D) $\left[\frac{1}{3}, \frac{13}{3}\right]$

Ans. B

Sol. -

- Q7. Let A and B be two events such that $P(\overline{A \cup B}) = \frac{1}{6}$, $P(A \cap B) = \frac{1}{4}$ and $P(\overline{A}) = \frac{1}{4}$, where $\overline{A}$ stands for complement of event A. Then events A and B are

- (A) equally likely and mutually exclusive (B) equally likely but not independent
(C) independent but not equally likely (D) mutually exclusive and independent

Ans. C

Sol. -

- Q8. Let E^c denote the complement of an event E. Let E, F, G be pairwise independent events with $P(G) > 0$ and $P(E \cap F \cap G) = 0$. Then $P(E^c \cap F^c | G)$ equals

- (A) $P(E^c) + P(F^c)$ (B) $P(E^c) - P(F^c)$ (C) $P(E^c) - P(F)$ (D) $P(E) - P(F^c)$

Ans. C

Sol. -

- Q9. If $P(A) = 0.25$, $P(B) = 0.50$ and $P(A \cap B) = 0.14$, then the probability that neither A nor B occurs is-

- (A) 0.39 (B) 0.25 (C) 0.11 (D) None of these

Ans. A

Sol. -

- Q10. A fair die is tossed. If the number is odd, find the probability that it is prime is

- (A) $\frac{2}{3}$ (B) $\frac{1}{2}$ (C) 1 (D) $\frac{1}{3}$

Ans. A

Sol. -

- Q11. A card is drawn from a well shuffled ordinary deck of 52-playing cards. The card drawn is found to be a spade. Then the probability that the card is an ace, is

- (A) $\frac{1}{13}$ (B) $\frac{1}{52}$ (C) $\frac{1}{4}$ (D) None of these

Ans. A

Sol. -

Q12. A pair of dice is thrown. If total of numbers turned up on both the dice is 8, then the probability that the number then up on the second die is 5' is

- (A) $\frac{5}{36}$ (B) $\frac{1}{6}$ (C) $\frac{1}{5}$ (D) $\frac{2}{5}$

Ans. C

Sol. -

Q13. A pair of numbers is picked up randomly (without replacement) from the set {1, 2, 3, 5, 7, 11, 12, 13, 17, 19}. The probability that the number 11 was picked given that the sum of the numbers was even, is nearly :

- (A) 0.1 (B) 0.125 (C) 0.24 (D) 0.18

Ans. C

Sol. -

Q14. Whenever horses a, b, c race together, their respective probabilities of winning the race are 0.3, 0.5 and 0.2 respectively. If they race three times the probability that "the same horse wins all the three races" and the probability that a, b, c each wins one race, are respectively (Assume no dead heat)

- (A) $\frac{8}{50}; \frac{9}{50}$ (B) $\frac{16}{50}; \frac{3}{50}$ (C) $\frac{12}{50}; \frac{15}{50}$ (D) $\frac{10}{50}; \frac{8}{50}$

Ans. A

Sol. -

Q15. An unbiased cubic die marked with 1, 2, 2, 3, 3, 3 is rolled 3 times. The probability of getting a total score of 4 or 6 is

- (A) $\frac{16}{216}$ (B) $\frac{50}{216}$ (C) $\frac{60}{216}$ (D) None

Ans. B

Sol. -

Q16. A problem in mathematics is given to three students A, B, C and their respectively probability of solving the problem is $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$. Probability that the problem is solved is.

- (A) $\frac{3}{4}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Ans. A

Sol. -

Q17. Let A & B be two events. Suppose $P(A) = 0.4$, $P(B) = p$ & $P(A \cup B) = 0.7$. The value of p for which A & B are independent is :

- (A) $1/3$ (B) $1/4$ (C) $1/2$ (D) $1/5$

Ans. C

Sol. -

Q18. Let two fair six-faced dice A and B be thrown simultaneously. If E_1 is the event that die A shows up four E_2 is the event that die B shows up two and E_3 is the event that the sum of numbers on both dice is odd, then which of the following statements is NOT true ?

- (A) E_1 and E_3 are independent (B) E_1, E_2 and E_3 are independent
(C) E_1 and E_2 are independent (D) E_2 and E_3 are independent

Ans. B

Sol. $P(E_1 \cap E_2 \cap E_3) = 0$
and $P(E_1) = 1/6$ $P(E_2) = 1/6$
 $P(E_3) = 1/2$
So E_1, E_2 & E_3 are not independent

Q19. An experiment has 10 equally likely outcomes. Let A and B be two non-empty events of the experiment. If A consists of 4 outcomes, the number of outcomes that B must have so that A and B are independent, is

- (A) 2, 4 or 8 (B) 3, 6, or 9 (C) 4 or 8 (D) 5 or 10

Ans. D

Sol. -

Q20. An unbiased die with the numbers 1, 2, 3, 4, 6 and 8 on its six faces is rolled. After this roll if an odd number appears on the top face, all odd numbers on the die are doubled. If an even number appears on the top face, all the even numbers are halved. If the given die changes in this way then the probability that the face 2 will appear on the second roll is

- (A) $2/18$ (B) $3/18$ (C) $2/9$ (D) $5/18$

Ans. C

Sol. -

Q21. A butterfly randomly lands on one of the six squares of the T-shaped figure shown and then randomly moves to an adjacent square. The probability that the butterfly ends up on the R square is

G			
Y	W	R	B
O			

- (A) $\frac{1}{4}$ (B) $\frac{1}{3}$ (C) $\frac{1}{5}$ (D) $\frac{1}{6}$

Ans. A

Sol. -

Q22. Three groups A, B, C are competing for positions of the Board of Directors of a company. The probabilities of their winning are 0.5, 0.3, 0.2 respectively. If the group A wins, the probability of introducing a new product is 0.7 and the corresponding probabilities for group B and C are 0.6 and 0.5 respectively. The probability that the new product will be introduced, is-

- (A) 0.18 (B) 0.35 (C) 0.10 (D) 0.63

Ans. D

Sol. -

Q23. A signal which can be green or red with probability $\frac{4}{5}$ and $\frac{1}{5}$ respectively, is received by station A and then transmitted to station B. The probability of each station receiving the signal correctly is $\frac{3}{4}$. If the signal received at station B is green, then the probability that the original signal was green is

- (A) $\frac{3}{5}$ (B) $\frac{6}{7}$ (C) $\frac{20}{23}$ (D) $\frac{9}{20}$

Ans. C

Sol. -

Q24. In a series of 3 independent trials the probability of exactly 2 success is 12 times as large as the probability of 3 successes. The probability of a success in each trial is:

- (A) $\frac{1}{5}$ (B) $\frac{2}{5}$ (C) $\frac{3}{5}$ (D) $\frac{4}{5}$

Ans. A

Sol. -

Q25. A bag contains 2 white & 4 black balls. A ball is drawn 5 times, each being replaced before another is drawn. The probability that at least 4 of the balls drawn are white is:

- (A) $\frac{4}{81}$ (B) $\frac{10}{243}$ (C) $\frac{11}{243}$ (D) none

Ans. C

Sol. -

Q26. A pair of fair dice is thrown independently three times. The probability of getting a score of exactly 9 twice is.

- (A) $\frac{8}{729}$ (B) $\frac{8}{243}$ (C) $\frac{1}{729}$ (D) $\frac{8}{9}$

Ans. B

Sol. -

Q27. The probability that an event A happens in one trial of an experiment is 0.4. Three independent trials of the experiment are performed. The probability that the event A happens at least once is-

- (A) 0.936 (B) 0.784 (C) 0.904 (D) None of these

Ans. B

Sol. -

Q28. If 12 identical balls are to be placed in 3 identical boxes, then the probability that one of the boxes contains exactly 3 balls is :

- (A) $\frac{55}{3} \left(\frac{2}{3}\right)^{11}$ (B) $55 \left(\frac{2}{3}\right)^{10}$ (C) $220 \left(\frac{1}{3}\right)^{12}$ (D) $22 \left(\frac{1}{3}\right)^{11}$

Ans. A

Sol. Binomial probability distribution. No of trials = 12; success: putting ball in that particular box; no of successes = 3, $p = 2/3$

$$\frac{{}^{12}C_3 \times 2^9}{3^{12}} = 1$$

Q29. A purse contains 2 six sided dice. One is a normal fair die, while the other has 2 ones, 2 threes, and 2 fives. A die is picked up and rolled. Because of some secret magnetic attraction of the unfair die, there is 75% chance of picking the unfair die and a 25% chance of picking a fair die. The die is rolled and shows up the face 3. The probability that a fair die was picked up, is

- (A) $\frac{1}{7}$ (B) $\frac{1}{4}$ (C) $\frac{1}{6}$ (D) $\frac{1}{24}$

Ans. A

Sol. -

Q30. On a Saturday night 20% of all drivers in U.S.A. are under the influence of alcohol. The probability that a driver under the influence of alcohol will have an accident is 0.001. The probability that a sober driver will have an accident is 0.0001. If a car on a saturday night smashed into a tree, the probability that the driver was under the influence of alcohol, is

- (A) 3/7 (B) 4/7 (C) 5/7 (D) 6/7

Ans. C

Sol. -

Q31. In a certain factory, machines A, B and C produce bolts. Of their production, machines A, B, and C produce 2%, 1% and 3% defective bolts respectively. Machine A produces 35% of the total output of bolts, machine B produces 25% and machine C produces 40%. A bolt is chosen at random from the factory's production and is found to be defective. The probability it was produced on machine C, is

- (A) $\frac{6}{11}$ (B) $\frac{23}{45}$ (C) $\frac{24}{43}$ (D) $\frac{3}{11}$

Ans. C

Sol. -

Q32. A bag contains 4 red and 6 black balls. A ball is drawn at random from the bag, its colour is observed and this ball along with two additional balls of the same colour are returned to the bag. If now a ball is drawn at random from the bag, then the probability that this drawn ball is red, is

- (A) $\frac{1}{5}$ (B) $\frac{3}{4}$ (C) $\frac{3}{10}$ (D) $\frac{2}{5}$

Ans. D

Sol. -

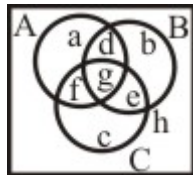
Q33. For three events A, B and C, $P(\text{Exactly one of A or B occurs}) = P(\text{Exactly one of B or C occurs}) = P(\text{Exactly one of C or A occurs}) = \frac{1}{4}$ and $P(\text{All the three events occur simultaneously}) = \frac{1}{16}$.

Then the probability that at least one of the events occurs, is

- (A) $\frac{7}{64}$ (B) $\frac{3}{16}$ (C) $\frac{7}{32}$ (D) $\frac{7}{16}$

Ans. D

Sol.



$$\begin{aligned} a + b + e + f &= \frac{1}{4} \\ b + d + c + f &= \frac{1}{4} \\ c + e + a + d &= \frac{1}{4} \end{aligned} \quad \text{Add}$$

$$g = \frac{1}{16}$$

$$2(a + b + c + d + e + f) = \frac{3}{4}$$

$$\text{Req. prob.} = a + b + c + d + e + f + g = \frac{3}{8} + \frac{1}{16} = \frac{7}{16}$$

Q34. From an urn containing 3 red & 2 white balls, a man is to draw 2 balls at random without replacement, being promised 20 paise for each red ball he draws & 10 paise for each white one. His expectation is:

- (A) 24 paise (B) 30 paise (C) 32 paise (D) 35 paise

Ans. C

Sol. -

Q35. A & B throw with one dice for a stake of Rs. 99/- which is to be won by the player who first throws 4. If A has the first throw then their respective expectations of rupees are:

- (A) 50 & 49 (B) 54 & 45 (C) 45 & 54 (D) none

Ans. B

Sol. -

Q36. Mean and variance of a Binomial variate are in the ratio of 3 : 2. The most probable number of happening of the variable in 10 trials of the experiment is

- (A) 2 (B) 3 (C) 4 (D) 5

Ans. B

Sol. -

Q37. A dice is tossed 5 times. Getting an odd number is considered a success. Then the variance of distribution of success is

- (A) $\frac{8}{3}$ (B) $\frac{3}{8}$ (C) $\frac{4}{5}$ (D) $\frac{5}{4}$

Ans. D

Sol. -

Q38. The mean and variance of a random variable X having binomial distribution are 4 and 2 respectively, then $P(X = 1)$ is

- (A) $\frac{1}{4}$ (B) $\frac{1}{32}$ (C) $\frac{1}{16}$ (D) $\frac{1}{8}$

Ans. B

Sol. -

Q39. A random variable X has the probability distribution:

X:	1	2	3	4	5	6	7	8
P(X):	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

For the events $E = \{X \text{ is a prime number}\}$ and $F = \{X < 4\}$, the probability $P(E \cup F)$ is.

- (A) 0.50 (B) 0.77 (C) 0.35 (D) 0.87

Ans. B

Sol. -

Q40. The mean and the variance of a binomial distribution are 4 and 2 respectively. Then the probability of 2 successes is.

- (A) $\frac{28}{256}$ (B) $\frac{219}{256}$ (C) $\frac{128}{256}$ (D) $\frac{37}{256}$

Ans. A

Sol. -

Q41. A box contains 15 green and 10 yellow balls. If 10 balls are randomly drawn, one-by-one, with replacement, then the variance of the number of green balls drawn is

- (A) 4 (B) $\frac{6}{25}$ (C) $\frac{12}{5}$ (D) 6

Ans. C

Sol. -

Exercise J02

Q1. A pair of fair dice is tossed. Find the probability that the maximum of the two numbers is greater than 4.

Ans. $5/9$

Sol. -

Q2. The chance of one event happening is the square of the chance of a 2nd event, but odds against the first are the cubes of the odds against the 2nd. Find the chances of each. (assume that both events are neither sure nor impossible).

Ans. $\frac{1}{9}, \frac{1}{3}$

Sol. -

Q3. In a given race, the odds in favour of four horses A, B, C & D are 1 : 3, 1 : 4, 1 : 5 and 1 : 6 respectively. Assuming that a dead heat is impossible, find the chance that one of them wins the race.

Ans. $319/420$

Sol. -

Q4. In a box, there are 8 alphabets cards with the letters: S, S, A, A, A, H, H, H. Find the probability that the word 'ASH' will form if:

(i) the three cards are drawn one by one & placed on the table in the same order that they are drawn.

(ii) the three cards are drawn simultaneously.

Ans. (i) $3/56$ (ii) $9/28$

Sol. -

Q5. A bag contains N balls, some of which are white, the others are black, white being more in number than black. Two balls are drawn at random from the bag, without replacement. It is found that the probability that the two balls are of the same colour is the same as the probability that they are of different colour.

It is given that $180 < N < 220$. If K denotes the number of white balls, find the exact value of $(K + N)$.

Ans. 301

Sol. -

Q6. In a batch of 10 articles, 4 articles are defective. 6 articles are taken from the batch for inspection. If more than 2 articles in this batch are defective, the whole batch is rejected. Find the probability that the batch will be rejected.

Ans. 19/42

Sol. -

Q7. A die is weighted such that the probability of rolling the face numbered n is proportional to n^2 ($n = 1, 2, 3, 4, 5, 6$). The die is rolled twice, yielding the numbers a and b . If the probability that $a < b$ can be expressed in lowest rational as $\frac{p}{q}$ find $(p + q)$.

Ans. 124

Sol. -

Q8. A covered basket of flowers has some lilies and roses. In search of rose, Sweety and Shweta alternately pick up a flower from the basket but puts it back if it is not a rose. Sweety is 3 times more likely to be the first one to pick a rose. If sweety begin this 'rose hunt' and if there are 60 lilies in the basket, find the number of roses in the basket.

Ans. 120

Sol. -

Q9. To pass a test a child has to perform successfully in two consecutive tasks, one easy and one hard task. The easy task he can perform successfully with probability 'e' and the hard task he can perform successfully with probability 'h', where $h < e$. He is allowed 3 attempts, either in the order (Easy, Hard, Easy) (option A) or in the order (Hard, Easy, Hard) (option B) whatever may be the order, he must be successful twice in a row. Assuming that his attempts are independent, in what order he chooses to take the tasks, in order to maximise his probability of passing the test.

Ans. B

Sol. -

Q10. The entries in a two-by-two determinant $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ are integers that are chosen randomly and independently, and, for each entry, the probability that the entry is odd is p . If the probability that the value of the determinant is even is $1/2$, then find the value of p .

Ans. $\frac{\sqrt{2}}{2}$

Sol. -

Q11. An aircraft gun can take a maximum of four shots at an enemy's plane moving away from it. The probability of hitting the plane at first, second, third & fourth shots are 0.4, 0.3, 0.2 & 0.1 respectively. What is the probability that the gun hits the plane .

Ans. 0.6976

Sol. -

Q12. If A and B are independent events, prove that $P(A \cup B) \cdot P(A' \cap B') \leq P(C)$, where C is an event defined that exactly one of A or B occurs.

Ans. -

Sol. -

Q13. There are 2 groups of subjects one of which consists of 5 science subjects & 3 engg. subjects & other consists of 3 science & 5 engg. subjects. An unbiased die is cast. If the number 3 or 5 turns up a subject is selected at random from first group, otherwise the subject is selected from 2nd group. Find the probability that an engg. subject is selected.

Ans. 13/24

Sol. -

Q14. There are 4 urns. The first urn contains 1 white & 1 black ball, the second urn contains 2 white & 3 black balls, the third urn contains 3 white & 5 black balls & the fourth urn contains 4 white & 7 black balls. The selection of each urn is not equally likely. The probability of selecting i^{th} urn is $\frac{i^2 + 1}{34}$ ($i = 1, 2, 3, 4$). If we randomly select one of the urns & draw a ball, then the probability of ball being white is $\frac{p}{q}$ where p and q $\in \mathbb{N}$ are in their lowest form. Find (p + q).

Ans. 2065

Sol. -

Q15. An author writes a good book with a probability of 1/2. If it is good it is published with a probability of 2/3. If it is not, it is published with a probability of 1/4. Find the probability that he will get at least one book published if he writes two books.

Ans. 407/576

Sol. -

Q16. There are 6 red balls and 6 green balls in a bag. Five balls are drawn out at random and placed in a red box. The remaining seven balls are put in a green box. If the probability that the number of red balls in the green box plus the number of green balls in the red box is not a prime number, is $\frac{p}{q}$ where p and q are relatively prime, then find the value of (p + q)

Ans. 37

Sol. -

Q17. A room has three electric lamps . From a collection of 10 electric bulbs of which 6 are good 3 are selected at random & put in the lamps. Find the probability that the room is lighted.

Ans. $\frac{29}{30}$

Sol. -

Q18. A bomber wants to destroy a bridge. Two bombs are sufficient to destroy it. If four bombs are dropped, what is the probability that it is destroyed, if the chance of a bomb hitting the target is 0.4.

Ans. $\frac{328}{625}$

Sol. -

Q19. A game is played with a special fair cubic die which has one red side, two blue sides, and three green sides. The result is the colour of the top side after the die has been rolled. If the die is rolled repeatedly, the probability that the second blue result occurs on or before the tenth roll, can be expressed in the form $\frac{3^p - 2^q}{3^r}$ where p, q, r are positive integers, find the value of $p^2 + q^2 + r^2$.

Ans. 283

Sol. -

Q20. A bag contains 12 red balls and 6 white balls. Six balls are drawn one by one without replacement of which atleast 4 balls are white. Find the probability that in the next two draws exactly one white ball is drawn (leave the answer in terms of nC_r).

Ans.
$$\frac{{}^{12}C_2 {}^6C_4 {}^{10}C_1 {}^2C_1 + {}^{12}C_1 {}^6C_5 {}^{11}C_1 {}^1C_1}{{}^{12}C_2 ({}^{12}C_2 {}^6C_4 + {}^{12}C_1 {}^6C_5 + {}^{12}C_0 {}^6C_6)}$$

Sol. -

Q21. A person goes to office either by car, scooter, bus or train the probability of which being $\frac{1}{7}$, $\frac{3}{7}$, $\frac{2}{7}$ and $\frac{1}{7}$ respectively. Probability that he reaches office late, if he takes car, scooter, bus or train is $\frac{2}{9}$, $\frac{1}{9}$, $\frac{4}{9}$ and $\frac{1}{9}$ respectively. Given that he reached office in time, then what is the probability that he travelled by a car.

Ans. $\frac{1}{7}$

Sol. -

Paragraph

There are n urns each containing n + 1 balls such that the i^{th} urn contains i white balls and (n + 1 – i) red balls. Let u_i be the event of selecting i^{th} urn, $i = 1, 2, 3, \dots, n$ and w denotes the event of getting a white ball.

\n

Q22. If $P(u_i) \propto i$ where $i = 1, 2, 3, \dots, n$ then $\lim_{x \rightarrow \infty} P(w)$ is equal to

- (A) 1 (B) $2/3$ (C) $3/4$ (D) $1/4$

Ans. B

Sol. -

Q23. If $P(u_i) = c$, where c is a constant then $P(u_n/w)$ is equal to

- (A) $\frac{2}{n+1}$ (B) $\frac{1}{n+1}$ (C) $\frac{n}{n+1}$ (D) $\frac{1}{2}$

Ans. A

Sol. -

Q24. If n is even and E denotes the event of choosing even numbered urn $P(u_i) = \frac{1}{n}$, then the value of $P(w/E)$, is

- (A) $\frac{n+2}{2n+1}$ (B) $\frac{n+2}{2(n+1)}$ (C) $\frac{n}{n+1}$ (D) $\frac{1}{n+1}$

Ans. B

Sol. -

Paragraph

Let U_1 and U_2 be two urns such that U_1 contains 3 white and 2 red balls, and U_2 contains only 1 white ball. A fair coin is tossed. If head appears then 1 ball is drawn at random from U_1 and put into U_2 . However, if tail appears then 2 balls are drawn at random from U_1 and put into U_2 . Now 1 ball is drawn at random from U_2 .

Q25. Given that the drawn ball from U_2 is white, the probability that head appeared on the coin is :-

- (A) $\frac{17}{23}$ (B) $\frac{11}{23}$ (C) $\frac{15}{23}$ (D) $\frac{12}{23}$

Ans. D

Sol. Required probability

$$\begin{aligned} &= \frac{2/5}{2/5 + 11/30} \text{ (Using Baye's theorem)} \\ &= \frac{12}{23} \end{aligned}$$

Q26. The probability of the drawn ball from U_2 being white is

- (A) $\frac{13}{30}$ (B) $\frac{23}{30}$ (C) $\frac{19}{30}$ (D) $\frac{11}{30}$

Ans. B

Sol. Required probability

$$= \frac{1}{2} \left(\frac{3}{5} \cdot 1 + \frac{2}{5} \cdot \frac{1}{2} \right) + \frac{1}{2} \left(\frac{{}^3C_2}{{}^5C_2} \cdot 1 + \frac{{}^2C_2}{{}^5C_2} \cdot \frac{1}{3} + \frac{{}^3C_1 {}^2C_1}{{}^5C_2} \cdot \frac{2}{3} \right)$$

$$= \frac{1}{4} \left(\frac{4}{5} \right) + \frac{1}{2} \left(\frac{3}{10} + \frac{1}{30} + \frac{2}{5} \right) = \frac{2}{5} + \frac{11}{30} = \frac{23}{30}$$

Q27. A student appears for tests I, II & III. The student is successful if he passes either in tests I & II or tests I & III. The probabilities of the student passing in the tests I, II & III are p, q &

1/2 respectively. If the probability that the student is successful is 1/2, then:

(A) $p = 1, q = 0$ (B) $p = 2/3, q = 1/2$ (C) $p = 3/5, q = 2/3$

(D) there are infinitely many values of p & q.

Ans. D, C, B, A

Sol. -

Q28. In an experimental performance of a single throw of a pair of unbiased normal dice, three events E_1, E_2 & E_3 are defined as follows:

E_1 : getting a prime numbered face on each dice

E_2 : getting the same number on each dice

E_3 : getting a sum total of dots on two dice equal to 8. Then:

(A) the events E_1, E_2 & E_3 are mutually exclusive

(B) the events E_1, E_2 & E_3 are not pairwise mutually exclusive

(C) the events E_1, E_2 are independent (D) $P(E_3|E_1) = 2/9$.

Ans. A, B, D

Sol. -

Q29. An unbiased coin is tossed n times. Let X denote the number of times head occurs. If $P(X = 4), P(X = 5)$ and $P(X = 6)$ are in AP, then the value of n can be

(A) 7 (B) 10 (C) 12 (D) 14

Ans. D, A

Sol. -

Q30. A fair coin is tossed repeatedly. If tail appears on first four tosses, then the probability of head appearing on fifth toss equals:

(A) 1/2 (B) 1/32 (C) 31/32 (D) 1/5

Ans. A

Sol. -

Q31. In throwing a die let A be the event 'coming up of an odd number', B be the event 'coming up of an even number', C be the event 'coming up of a number ≥ 4 ' and D be the event 'coming up of a number < 3 ', then

- (A) A and B are mutually exclusive and exhaustive
- (B) A and C are mutually exclusive and exhaustive
- (C) A, C and D form an exhaustive system
- (D) B, C and D form an exhaustive system

Ans. C, A

Sol. -

Q32. Let $0 < P(A) < 1$, $0 < P(B) < 1$ & $P(A \cup B) = P(A) + P(B) - P(A) \cdot P(B)$, then:

- (A) $P(B/A) = P(B) - P(A)$ (B) $P(A^C \cup B^C) = P(A^C) + P(B^C)$
- (C) $P((A \cup B)^C) = P(A^C) \cdot P(B^C)$ (D) $P(A/B) = P(A)$

Ans. C, D

Sol. -

Q33. A fair coin is tossed 99 times. If X is the number of times heads occur, then $P(X = r)$ is maximum when r is

- (A) 49 (B) 50 (C) 51 (D) None of these

Ans. A, B

Sol. -

Q34. A manufacturer of airplane parts makes a certain engine that has a probability p of failing on any given flight. There are two planes fitted with this type of engine. One plane has 3 such engines and other plane has 5. A plane crashes if more than half the engines fitted in it fail. If the two plane models have the same probability of crashing then the value of p can be

- (A) 0 (B) 1 (C) $1/2$ (D) $3/5$

Ans. A

Sol. -

Q35. Let E and F be two independent events. The probability that exactly one of them occurs is $\frac{11}{25}$ and the probability of none of them occurring is $\frac{2}{25}$. If P(T) denotes the probability of occurrence of the event T, then

- (A) $P(E) = \frac{4}{5}$, $P(F) = \frac{3}{5}$ (B) $P(E) = \frac{1}{5}$, $P(F) = \frac{2}{5}$ (C) $P(E) = \frac{2}{5}$, $P(F) = \frac{1}{5}$

$$^{(D)} P(E) = \frac{3}{5}, P(F) = \frac{4}{5}$$

Ans. A, D

Sol. -

Q36. A normal coin is tossed four times. Two event E and F are defined as

E : no two consecutive heads occur in 4 tosses.

F : At least 2 consecutive heads occur in 4 tosses.

The events E and F are :

^(A) equally likely ^(B) mutually exclusive ^(C) exhaustive

^(D) such that one is twice as likely to occur as other

Ans. A, B, C

Sol. $P(E) = P(F) = \frac{1}{2} \Rightarrow A, B, C$

$$\left. \begin{array}{ll} \text{H T H H} & \text{H T T H} \\ \text{T H H H} & \text{T H T H} \\ \text{T T H H} & \text{T T T H} \\ \text{H H H H} & \text{H H T H} \end{array} \right\} \left\{ \begin{array}{ll} \text{H T H T} & \text{H T T T} \\ \text{T H H T} & \text{T H T T} \\ \text{T T H T} & \text{T T T T} \\ \text{H H H T} & \text{H H T T} \end{array} \right.$$

$$P(E) = \frac{8}{16} = \frac{1}{2} = P(F)$$

Q37. For two given events A and B, $P(A \cap B)$ is :

^(A) not less than $P(A) + P(B) - 1$ ^(B) not greater than $P(A) + P(B)$

^(C) equal to $P(A) + P(B) - P(A \cup B)$ ^(D) equal to $P(A) + P(B) + P(A \cup B)$

Ans. A, B, C

Sol. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$0 \leq P(A \cup B) \leq 1$$

$$P(A) + P(B) - 1 \leq P(A \cap B) \leq P(A) + P(B)$$

Q38. For $P(A) = \frac{3}{8}$; $P(B) = \frac{1}{2}$; $P(A \cup B) = \frac{5}{8}$ which of the following do/does hold good?

^(A) $P(A^c/B) = 2P(A/B^c)$ ^(B) $P(B) = P(A/B)$

^(C) $15 P(A^c/B^c) = 8 P(B/A^c)$ ^(D) $P(A/B^c) = P(A \cap B)$

Ans. A, B, D

Sol. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\frac{5}{8} = \frac{3}{8} + \frac{4}{8} - P(A \cap B) \Rightarrow P(A \cap B) = \frac{2}{8} = \frac{1}{4}$$

$$\text{Now } P(A^c/B) = \frac{P(A^c \cap B)}{P(B)} = \frac{P(B) - P(A \cap B)}{P(B)} = 1 - \frac{2}{8} \cdot \frac{8}{4} = \frac{1}{2}$$

$$\text{Now, } 2P(A/B^c) = \frac{2P(A \cap B^c)}{P(B^c)} = \frac{2(P(A) - P(A \cap B))}{1 - P(B)}$$

$$= 4 \left(\frac{3}{8} - \frac{2}{8} \right) = \frac{1}{2} \Rightarrow (A) \text{ is correct}$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1}{4} \cdot \frac{2}{1} = \frac{1}{2} \Rightarrow (B) \text{ is correct}$$

$$\text{again } P(A^c/B^c) = \frac{P(A^c \cap B^c)}{P(B^c)} = \frac{1 - P(A \cup B)}{1 - P(B)} = 2 \left(1 - \frac{5}{8} \right) = \frac{3}{4}$$

$$P(B/A^c) = \frac{P(B \cap A^c)}{1 - P(A)} = \frac{P(B) - P(A \cap B)}{5/8} = \frac{\frac{1}{2} - \frac{1}{4}}{\frac{5}{8}} = \frac{1}{4} \cdot \frac{8}{5} = \frac{2}{5}$$

$$\text{Hence } 8P(A^c/B^c) = 15P(B/A^c) \Rightarrow (C) \text{ is not correct}$$

$$\text{again } 2P(A/B^c) = \frac{1}{2} \text{ from (1)} \Rightarrow P(A/B^c) = \frac{1}{4} = P(A \cap B)$$

hence (D) is correct

Q39. Which of the following statement(s) is/are correct?

(A) 3 coins are tossed once. Two of them atleast must land the same way. No matter whether they land heads or tails, the third coin is equally likely to land either the same way or oppositely. So, the chance that all the three coins land the same way is $1/2$.

(B) Let $0 < P(B) < 1$ and $P(A/B) = P(A/B^c)$ then A and B are independent.

(C) Suppose an urn contains 'w' white and 'b' black balls and a ball is drawn from it and is replaced along with 'd' additional balls of the same colour. Now a second ball is drawn from it. The probability that the second drawn ball is white is independent of the value of 'd'.

(D) A, B, C simultaneously satisfy $P(ABC) = P(A) \cdot P(B) \cdot P(C)$ and $P(AB\bar{C}) = P(A) \cdot P(B) \cdot P(\bar{C})$ and $P(A\bar{B}C) = P(A) \cdot P(\bar{B}) \cdot P(C)$ and $P(\bar{A}BC) = P(\bar{A}) \cdot P(B) \cdot P(C)$ then A, B, C are independent.

Ans. B, C, D

Sol.

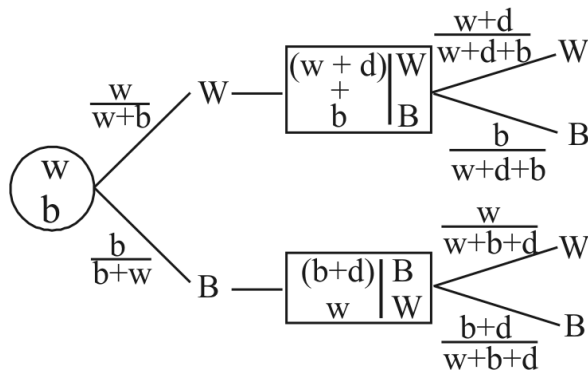
(A) False; $P(TTT \text{ or } HHH) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$

(B) $\frac{P(A \cap B)}{P(B)} = \frac{P(A \cap B^c)}{1 - P(B)} = \frac{P(A) - P(A \cap B)}{1 - P(B)}$

$P(A \cap B)[1 - P(B)] = P(B) \cdot P(A) - P(B) \cdot P(A \cap B)$

$P(A \cap B) = P(A) \cdot P(B) \Rightarrow \text{True}$

(C)



$P(W)$

$$\frac{w}{w+b} \cdot \frac{w+d}{w+d+b} + \frac{b}{b+w} \cdot \frac{w}{w+b+d} = \frac{w}{(b+w)(w+d+b)}(w+d+b)$$

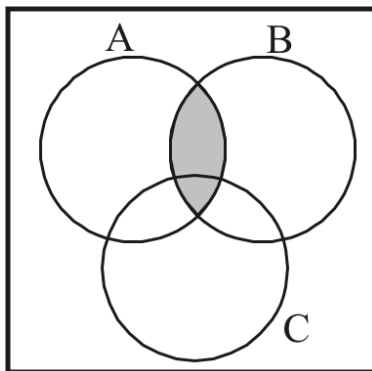
Hence the probability that 2nd ball drawn is white is independent of 'd'.

(D) To prove that A, B, C are pairwise independent also

now $P(A \cap B) = P(A \cap B \cap \bar{C} \cup A \cap B \cap C)$ (from the venn diagram)

$P(A \cap B) = P(A \cap B \cap \bar{C}) + P(A \cap B \cap C)$

$= P(A) \cdot P(B) \cdot P(C) + P(A) \cdot P(B) \cdot P(C)$ (given)



$= P(A) \cdot P(B) [P(C) + P(\bar{C})]$

$= P(A) \cdot P(B)$

|||ly for other two $\Rightarrow$ (D) is correct

Q40. Two fair dice are thrown simultaneously. 3 events A, B and C are defined as follows :

A : odd face on first dice ;

B : odd face with second dice

C : sum of the points on two dice is odd.

Then :

- (A) the events A, B and C are mutually exclusive
- (B) the events A, B and C are mutually independent
- (C) the events A, B and C are pairwise independent
- (D) $P(A \cap B) = P(B \cap C) = P(C \cap A)$

Ans. C, D

Sol. A = odd face on first dice

$$A = \left\{ \begin{array}{l} (1,1), (1,2), (1,3), (1,4), (1,5), (1,6) \\ (3,1), (3,2), (3,3), (3,4), (3,5), (3,6) \\ (5,1), (5,2), (5,3), (5,4), (5,5), (5,6) \end{array} \right\}$$

B = odd face with second dice

$$B = \left\{ \begin{array}{l} (1,1), (1,3), (1,5), (1,6), (1,2), (1,4) \\ (3,1), (3,3), (3,5), (3,6), (3,2), (3,4) \\ (5,1), (5,3), (5,5), (5,6), (5,2), (5,4) \end{array} \right\}$$

C = sum of the points on two dice is odd

$$C = \left\{ \begin{array}{l} (1,2), (1,4), (1,6), (2,1), (2,3), (2,5) \\ (3,2), (3,4), (3,6), (4,1), (4,3), (4,5) \\ (5,2), (5,4), (5,6), (6,1), (6,3), (6,5) \end{array} \right\}$$

Q41. Two events A and B are such that the probability that at least one of them occurs is $5/6$ and both of them occurring simultaneously is $1/3$. If the probability of not occurrence of B is $1/2$ then :

- (A) A and B are equally likely
- (B) A and B are independent
- (C) $P(A/B) = 2/3$
- (D) $3 P(A) = 4 P(B)$

Ans. B, C, D

Sol. $P(A \cup B) = \frac{5}{6}$; $P(A \cap B) = \frac{1}{3}$; $P(B) = \frac{1}{2}$
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\frac{5}{6} = P(A) + \frac{1}{2} - \frac{1}{3} \Rightarrow P(A) = \frac{2}{3}$$

$$P(A \cap B) = \frac{1}{3} = P(A) \cdot P(B) \Rightarrow (B)$$

$$\text{hence } P(A/B) = P(A) = \frac{2}{3} \Rightarrow (C)$$

$$\text{Also } \frac{P(B)}{P(A)} = \frac{1}{2} \times \frac{3}{2} = \frac{3}{4} \Rightarrow 3P(A) = 4P(B) \Rightarrow (D)$$

Q42. The probabilities of events, $A \cap B$, A , B & $A \cup B$ are respectively in A.P. with probability of second term equal to the common difference. Therefore the events A and B are :

- (A) mutually exclusive (B) independent (C) such that one of them must occur
(D) such that one is twice as likely as the other

Ans. A, D

Sol. $P(A \cap B)$, $P(A)$, $P(B)$, $P(A \cup B)$ are in A.P. with $d = P(A)$

$$\therefore P(A) - P(A \cap B) = P(A) \Rightarrow P(A \cap B) = 0 \Rightarrow A \text{ \& B are ME}$$

$$\text{also } P(B) - P(A) = P(A) \Rightarrow 2P(A) = P(B)$$

$\therefore$ if $P(A) = p$; $P(B) = 2p \Rightarrow (D)$ compatible means which can happen simultaneously

Q43. Which of the following statements is/are True?

- (A) A fair coin is tossed n times where n is a positive integer. The probability that n^{th} toss results in head is $1/2$.
(B) The conditional probability that the n^{th} toss results in head given that first $(n - 1)$ tosses results in head is $1/2^n$
(C) Let E and F be the events such that F is neither impossible nor sure. If $P(E/F) > P(E)$ then $P(E/F^c) > P(E)$
(D) If A , B and C are independent then the events $(A \cup B)$ and C are independent.

Ans. A, D

Sol. (D) to prove that $P(C \cap (A \cup B)) = P(C) \cdot P(A \cup B)$

$$P((C \cap A) + (C \cap B))$$

$$= P(C \cap A) + P(C \cap B) - P(A \cap B \cap C)$$

$$= P(C) \cdot P(A) + P(C) \cdot P(B) - P(A) \cdot P(B) \cdot P(C)$$

$$= P(C) \cdot [P(A) + P(B) - P(A \cap B)]$$

$$= P(C) \cdot P(A \cup B) \Rightarrow C \text{ and } A \cup B \text{ are independent}$$

Q44.	Column – I	Column –
	II	
	(A) If the probability that units digit in square of an even integer is 4 is p, then the value of 5p is	(P) 1
	(B) If A and B are independent events and $P(A \cap B) = \frac{1}{6}$, $P(\bar{A}) = \frac{2}{3}$, then $6P(B/A) =$	(Q) 2
	(C) Among 2 children, a child may equally be a boy or a girl if the probability that exactly one of them is a boy is p, then 6p =	(R) 3
	(D) A boy has 20% chance of hitting at a target. Let p denote the probability of hitting the target for the first time at the nth trial. If p satisfies the inequality $625p^2 - 175p + 12 < 0$, then value of n is	(S) 4
	(A) $(A) \rightarrow (Q), (B) \rightarrow (R), (C) \rightarrow (R), (D) \rightarrow (R)$	
	(B) $(A) \rightarrow (R), (B) \rightarrow (Q), (C) \rightarrow (Q), (D) \rightarrow (R)$	
	(C) $(A) \rightarrow (S), (B) \rightarrow (Q), (C) \rightarrow (R), (D) \rightarrow (S)$	
	(D) $(A) \rightarrow (Q), (B) \rightarrow (R), (C) \rightarrow (S), (D) \rightarrow (Q)$	

Ans. 5ef441569fff15766e2e099e

Sol. -

Q45.	Column – I	Column – II
	(A) A pair of dice is thrown. If total of numbers turned up on both the dies is 8, then the probability that the number then up on the second die is 5' is	(P) $\frac{5}{16}$
	(B) A box contains 4 white and 3 black balls. Two balls are drawn successively and is found that second ball is white, then the probability that Ist ball is also white is	(Q) $\frac{1}{3}$
	(C) A biased coin with probability p, $0 < p < 1$ of heads is tossed until a head appears for the first time. If the probability that the number of tosses required is even is $\frac{2}{5}$, then p equals	(R) $\frac{1}{2}$
	(D) A coin whose faces are marked 3 and 5 is tossed 4 times : what	(S) $\frac{1}{5}$

is the probability that the sum of the numbers thrown being less,
than 15?

(A) $(A) \rightarrow (P), (B) \rightarrow (R), (C) \rightarrow (Q), (D) \rightarrow (S)$

(B) $(A) \rightarrow (S), (B) \rightarrow (R), (C) \rightarrow (Q), (D) \rightarrow (P)$

(C) $(A) \rightarrow (S), (B) \rightarrow (Q), (C) \rightarrow (R), (D) \rightarrow (P)$

(D) $(A) \rightarrow (P), (B) \rightarrow (Q), (C) \rightarrow (R), (D) \rightarrow (S)$

Ans. 5ef443489fff15766e2e09d5

Sol. -